

# The $U(3)$ Gelfand–Tsetlin System: Exact Fibers, Linear Dynamics, and Unshifted Quantization

HCS-C365 theorem package

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## Abstract

We close the classical and quantum Gelfand–Tsetlin descriptions for a Hermitian  $U(3)$  coadjoint orbit under explicit conventions. An arrow completion proves the exact interlacing image; nested-group flows give every regular Lagrangian torus; character duality classifies every linear Hamiltonian closure and least period. With integral highest weight and no half-form correction, branching gives the complete basis with unshifted GT labels and the Weyl count. Repeated weights, facets, and nonintegral spectra are separated sharply. Exact enumeration is a regression receipt, not the proof.

## 1 Frozen orbit and exact theorem

Let  $\lambda_1 \geq \lambda_2 \geq \lambda_3$ , and let  $\mathcal{O}_\lambda$  be the orbit of Hermitian matrices with this spectrum. For skew-Hermitian  $X, Y$ , freeze

$$\omega_A([X, A], [Y, A]) = -i \operatorname{Tr}(A[X, Y]), \quad f_X(A) = -i \operatorname{Tr}(AX), \quad \iota_{X_{f_X}} \omega = df_X.$$

Thus  $X_{f_X}(A) = [X, A]$ . Write  $B = A^{(2)}$  for the northwest  $2 \times 2$  minor, let  $y_1 \geq y_2$  be its eigenvalues, and put  $x = A_{11}$ .

**Theorem 1** *The map  $F = (y_1, y_2, x)$  has image*

$$\Delta_\lambda = \{\lambda_1 \geq y_1 \geq \lambda_2 \geq y_2 \geq \lambda_3, \quad y_1 \geq x \geq y_2\}.$$

*If  $\lambda_1 > \lambda_2 > \lambda_3$ , every strict-interlacing fiber is exactly an effective Lagrangian  $T^3$ -orbit, with each circle normalized to period  $2\pi$ .*

*For  $H_\nu = \nu_1 y_1 + \nu_2 y_2 + \nu_3 x$ , its regular flow is  $\theta(t) = \theta_0 + t\nu \pmod{2\pi}$ . Let  $K_\nu$  be the subgroup annihilated by every  $k \in \mathbb{Z}^3$  satisfying  $k \cdot \nu = 0$ . The orbit closure is the coset  $\theta_0 + K_\nu$ , and  $\theta_0 + K_\nu = K_\nu$  exactly when  $\theta_0 \in K_\nu$ , with  $\dim K_\nu = \operatorname{rank}_{\mathbb{Q}}\{\nu_1, \nu_2, \nu_3\}$ . It is fixed for  $\nu = 0$ . For  $\nu \neq 0$ , it is periodic precisely when  $\nu = \alpha m$ , where  $\alpha > 0$  and  $m \in \mathbb{Z}^3$  is primitive; its least period is  $2\pi/\alpha$ .*

*If  $\lambda \in \mathbb{Z}^3$  is dominant, take prequantum curvature  $-i\omega$  and no half-form correction. Borel–Weil quantization has a multiplicity-free GT branching basis indexed, without a  $\rho$ -shift, by*

$$\lambda_1 \geq m_{21} \geq \lambda_2 \geq m_{22} \geq \lambda_3, \quad m_{21} \geq m_{11} \geq m_{22},$$

*with joint branching labels  $(m_{21}, m_{22}, m_{11})$ , and*

$$\dim Q(\mathcal{O}_\lambda) = \frac{(\lambda_1 - \lambda_2 + 1)(\lambda_2 - \lambda_3 + 1)(\lambda_1 - \lambda_3 + 2)}{2}.$$

## 2 Interlacing and constructive surjectivity

Cauchy interlacing gives all six inequalities. Conversely, at any point of  $\Delta_\lambda$ , set

$$B_{11} = x, \quad B_{12} = B_{21} = r, \quad B_{22} = y_1 + y_2 - x, \quad r^2 = (y_1 - x)(x - y_2).$$

This has eigenvalues  $y_1, y_2$ . In a basis diagonalizing  $B$ , consider

$$A'_{11} = y_1, \quad A'_{22} = y_2, \quad A'_{12} = 0, \quad A'_{13} = z_1, \quad A'_{23} = z_2, \quad A'_{33} = c = \lambda_1 + \lambda_2 + \lambda_3 - y_1 - y_2,$$

with the lower-triangular entries fixed by Hermitian symmetry. If  $P(t) = \prod_{j=1}^3 (t - \lambda_j)$ , choose

$$|z_1|^2 = -\frac{P(y_1)}{y_1 - y_2}, \quad |z_2|^2 = \frac{P(y_2)}{y_1 - y_2}.$$

Interlacing makes both quantities nonnegative. The monic cubic  $\det(tI - A')$  has the correct trace and agrees with  $P$  at  $y_1, y_2$ , hence equals  $P$ . Conjugating the first two coordinates back retains  $A_{11} = x$ . Endpoint coincidences follow by setting the vanishing arrow coordinate to zero, or by continuity. This proves the exact image.

### 3 Thimm flows and complete linear dynamics

Where  $B$  is simple,  $dy_j$  is its spectral projector in the northwest block. The Hamiltonian flow is conjugation by  $e^{itP_j}$ , while the  $x$ -flow is conjugation by  $e^{itE_{11}}$ . Each has period  $2\pi$ . Spectral functions on the same  $U(2)$  block commute; invariance along the nested  $U(1) \subset U(2)$  action gives commutation with  $x$ . Strict interlacing makes the three infinitesimal actions independent.

For completeness, a regular fiber has no hidden component. In the arrow construction, strictness makes  $r, z_1, z_2$  nonzero. After fixing an eigenvector gauge, their three independent phase choices form one  $T^3$  and exhaust every matrix having the frozen  $x, y_1, y_2$ ; changing the gauge merely relabels these phases. Thus the fiber is precisely the compact three-dimensional orbit. It is half-dimensional and isotropic, hence Lagrangian.

The linear-flow assertion now follows on  $\mathbb{R}^3/(2\pi\mathbb{Z})^3$ . The closure through the origin is the compact subgroup  $K_\nu$  annihilated by  $\{k \in \mathbb{Z}^3 : k \cdot \nu = 0\}$ . Character duality gives  $\dim K_\nu = \text{rank}_{\mathbb{Q}}\{\nu_j\}$ . Translating by  $\theta_0$  gives the general closure  $\theta_0 + K_\nu$ , which equals  $K_\nu$  if and only if  $\theta_0 \in K_\nu$ . Thus a nonzero basepoint can still give the subgroup: for rational rank three  $K_\nu = T^3$ , while for  $\nu = (1, 2, 3)$  the point  $(\pi/2, \pi, 3\pi/2)$  lies in  $K_\nu$ . By contrast  $(0, 0, \pi)$  does not. Rank one gives  $\nu = \alpha m$  with  $m$  primitive, and  $t\nu \in 2\pi\mathbb{Z}^3$  first at  $t = 2\pi/\alpha$ .

### 4 Boundary atlas

For a regular spectrum, every polytope facet is singular: at least one circle stabilizer occurs, and no universal topology is claimed at facet intersections. If  $\lambda_1 = \lambda_2 > \lambda_3$ , then  $y_1 = \lambda_1$ ; if  $\lambda_1 > \lambda_2 = \lambda_3$ , then  $y_2 = \lambda_3$ . In either case the orbit is four-dimensional, the image is a triangle, and only its regular interior carries the effective  $T^2$  statement. A scalar spectrum gives a point. A nonintegral ordered spectrum retains the classical theorem but is outside the quantization assertion.